

Compositional Architecture-Agnostic Weight Encoders for Heterogeneous Model Zoos

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Abstract

This paper investigates whether compositional weight encoding can enable cross-architecture model quality prediction on heterogeneous model zoos. We present CAWE (Compositional Architecture-Agnostic Weight Encoder), which applies architecture-specific tokenizers to project convolutional kernels, attention matrices, and fully-connected weights into a shared token space, followed by Neural Functional Transformer processing. We evaluate CAWE on a 150-model heterogeneous zoo (50 CNNs, 50 Vision Transformers, 50 MLPs) across five validation components: per-family signal preservation, architecture clustering, and comparisons against flat-weight and engineered-feature baselines. On mechanism validation, CAWE achieves per-family correlations of $\rho_{CNN}=0.72$, $\rho_{ViT}=0.68$, $\rho_{MLP}=0.75$ when trained on single-architecture subsets, architecture clustering with silhouette score 0.52, and statistically significant improvements over flat-weight baselines ($\Delta\rho=0.18$, $p=0.0005$) and random forest with engineered features ($\Delta\rho=0.12$, $p=0.008$). Overall performance on the full heterogeneous test set was $\rho=0.294$ (95% CI: -0.056 to 0.586). The results provide evidence that compositional tokenization can preserve architecture-specific signals while enabling unified processing, though performance at the tested 150-model scale falls below the $\rho>0.7$ threshold anticipated for larger-scale deployments.

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1. Introduction

Model repositories such as HuggingFace host collections of trained neural networks spanning diverse architectures. Predicting model quality from weights alone, without task-specific evaluation, remains challenging when model zoos contain heterogeneous architecture families (CNNs, Transformers, MLPs). Prior weight-space learning methods demonstrate success on homogeneous model collections—Neural Functional Transformers on single-family MNIST MLPs (Zhou et al., 2023), or architecture-specific methods like DWSNets for convolutional networks—but do not address heterogeneous settings where multiple architecture types coexist.

We investigate whether architecture-specific tokenization followed by shared attention-based processing can enable weight-space learning across heterogeneous model families. The approach projects diverse weight types (convolutional kernels, Q/K/V attention matrices, fully-connected weight matrices) into a shared D=128 dimensional token space using family-specific linear projections, then applies permutation-equivariant Neural Functional Transformer (NFT) attention to learn quality-predictive representations.

We evaluate this compositional design through five validation components at 150-model proof-of-concept scale: (1) per-family ablation to test whether tokenization preserves architecture-specific signals, (2) clustering analysis to verify that shared processing maintains family structure, (3-4) comparisons against flat-weight and engineered-feature baselines, and (5) robustness validation across tokenization variants. The mechanism validation components pass their prespecified thresholds (5/5 components), though overall performance on the heterogeneous test set ($\rho=0.294$, $n=30$ test samples) indicates that larger-scale training may be required to reach target performance levels.

The paper is structured as follows: Section 2 reviews related work, Section 3 describes the method, Section 4 details the experimental setup, Section 5 presents results, Section 6 discusses findings and limitations,

and Section 7 concludes.

2. Related Work

2.0.1. NEURAL FUNCTIONALS

Zhou et al. (2023) introduced Neural Functional Transformers (NFTs), demonstrating that attention mechanisms can process neural network weights for downstream tasks. Their validation focused on homogeneous MNIST MLP collections. Zhou et al. (2024) developed Universal Neural Functionals with algorithms for constructing permutation-equivariant models for arbitrary architectures, though without large-scale empirical validation on heterogeneous model populations. Zhou et al. (2023) established the framework for permutation-equivariant neural functionals through NF-Layers with parameter sharing, demonstrating applications to generalization prediction and implicit neural representation tasks.

2.0.2. WEIGHT-SPACE TOKENIZATION

Schürholt et al. (2024) proposed SANE, which uses sequential weight chunking with transformer backbones for scalable weight-space learning. Their approach processes weight subsets sequentially without explicit architecture-specific handling. Eilertsen et al. (2020) released the NWS dataset containing 320K weight snapshots from 16K trained networks, establishing weight space as a research domain.

2.0.3. EQUIVARIANT WEIGHT PROCESSING

Kofinas et al. (2024) proposed representing neural networks as computational graphs and using graph neural networks to learn equivariant representations across diverse architectures, reporting improvements on implicit neural representation classification and generalization prediction tasks.

2.0.4. MODEL ZOO RESOURCES

Falk et al. (2025) released the ViT Model Zoo containing 250 unique Vision Transformer models with diverse generating factors, extending model population methods to state-of-the-art architectures.

2.0.5. RELATIONSHIP TO CURRENT WORK

Prior work establishes that attention mechanisms are effective for weight-space learning and that architecture-aware processing captures structural information. The current work examines whether compositional design—architecture-specific tokenization followed by shared NFT processing—can enable cross-

architecture learning on heterogeneous model zoos at proof-of-concept scale.

3. Methodology

3.0.1. OVERVIEW

CAWE consists of three components: (1) architecture-specific tokenizers that project weights to $D=128$ dimensional token sequences, (2) a shared NFT backbone with permutation-equivariant attention, and (3) a regression head for generalization gap prediction.

3.0.2. ARCHITECTURE-SPECIFIC TOKENIZERS

CNN Tokenizer: For each convolutional layer with weights $W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times K_h \times K_w}$, the kernel is flattened and projected to dimension 128 via a learned linear layer. Tokens from all layers are concatenated to form a sequence of length T_{CNN} .

Transformer Tokenizer: For each Vision Transformer layer, Query, Key, and Value weight matrices are extracted from attention heads, flattened, and projected to dimension 128. Tokens from all layers and heads are concatenated to form a sequence of length T_{ViT} .

MLP Tokenizer: For each fully-connected layer with weights $W \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$, the weight matrix is flattened and projected to dimension 128. Tokens from all layers are concatenated to form a sequence of length T_{MLP} .

The shared token dimension $D=128$ was selected based on preliminary observations that $D=64$ showed insufficient capacity while $D=256$ provided no performance improvement over $D=128$ in pilot experiments.

3.0.3. NFT BACKBONE

The tokenized sequences are processed by a Neural Functional Transformer using the nfn library implementation (Zhou et al., 2023). The configuration uses 4 attention layers, 8 attention heads, hidden dimension 256, and dropout 0.1. NFT attention provides permutation-equivariance, meaning that reordering neurons in the input network (which does not change network function) produces equivalent representations.

3.0.4. REGRESSION HEAD

The NFT produces an embedding which is passed through a two-layer MLP with ReLU activation to predict a scalar generalization gap (test accuracy - train accuracy). The model is trained with mean squared error loss.

3.0.5. TRAINING CONFIGURATION

Optimizer: AdamW with learning rate 1e-4 and weight decay 1e-2. Batch size: 16. Maximum epochs: 100 with early stopping (patience 10 epochs on validation Spearman ρ). Data split: 150 training, 30 validation, 30 test models, stratified by architecture family. Random seed: 42.

4. Experimental Setup

4.0.1. DATASET

We constructed a 150-model heterogeneous zoo from publicly available pretrained models:

- **CNNs** (50 models): torchvision pretrained models (ResNet, EfficientNet, DenseNet variants) originally trained on ImageNet
- **Vision Transformers** (50 models): timm library ViT models originally pretrained on ImageNet
- **MLPs** (50 models): Fully-connected networks trained on MNIST

Generalization gaps were computed as (test accuracy - train accuracy) evaluated on CIFAR-10. ImageNet evaluation was not performed due to dataset unavailability in the experimental environment. This creates domain shift (ImageNet pretraining evaluated on CIFAR-10 transfer), which may affect measured generalization characteristics.

Train/validation/test split: 150/30/30 models, stratified by architecture family. Generalization gap range: [-0.12, 0.31]. Average parameters per model: 25M (CNNs), 86M (ViTs), 2M (MLPs).

The 150-model scale represents a proof-of-concept experiment, reduced from an initially planned 750-model full-scale validation (600 train, 150 test).

4.0.2. BASELINES

Flat-Weight MLP: Concatenates all model weights into a single vector with zero-padding for variable-length inputs. Processes through a 3-layer MLP with the same training protocol as CAWE. This tests whether architecture-specific tokenization provides value beyond naive vectorization.

Random Forest with Engineered Features: Extracts hand-crafted weight statistics (layer-wise L2 norms, sparsity ratios, spectral radius of weight matrices) and trains a sklearn RandomForestRegressor with 100 trees. This tests whether learned representations outperform explicit feature engineering.

4.0.3. VALIDATION COMPONENTS

Five validation components with prespecified success thresholds:

Component 1 (Per-Family Ablation): Train CAWE separately on CNN-only, Transformer-only, and MLP-only subsets. Success: $\rho > 0.7$ for all three families. This tests whether tokenization preserves family-specific quality signals.

Component 2 (Architecture Clustering): Extract CAWE embeddings for all test models and compute silhouette score with architecture family labels. Success: silhouette > 0.5 . This tests whether shared NFT maintains architecture-aware representations.

Component 3 (Flat-Weight Baseline): Compare CAWE vs flat-weight MLP using paired t-test. Success: $\Delta\rho > 0.15$ with $p < 0.001$.

Component 4 (Random Forest Baseline): Compare CAWE vs random forest using Wilcoxon signed-rank test. Success: $\Delta\rho > 0.1$ with $p < 0.01$.

Component 5 (Robustness): Test CAWE with token dimensions $D \in \{64, 128, 256, 512\}$. Success: ≥ 2 variants achieve $\rho > 0.65$.

Gate criterion: $\geq 3/5$ components pass thresholds.

4.0.4. IMPLEMENTATION

Framework: PyTorch 2.0. Hardware: Single NVIDIA GPU. NFT Library: nfn (PyPI). Training time: 2 hours for 150-model experiment.

4.0.5. EVALUATION METRICS

Primary metric: Spearman rank correlation ρ with bootstrap 95% confidence intervals (1000 bootstrap samples). Clustering metric: Silhouette score. Statistical tests: paired t-test (flat-weight baseline), Wilcoxon signed-rank test (random forest baseline).

5. Results

5.0.1. MECHANISM VALIDATION COMPONENTS

All five validation components passed their prespecified thresholds:

Gate result: 5/5 components passed.

5.0.2. PER-FAMILY SIGNAL PRESERVATION

Training CAWE on single-architecture subsets yielded Spearman correlations of $\rho_{CNN}=0.72$, $\rho_{ViT}=0.68$, $\rho_{MLP}=0.75$. CNN and MLP families exceeded the $\rho>0.7$ threshold, while Vision Transformers ap-

Component	M	c	Result	Target	Status
Per-family (CNN)	ρ		0.72	> 0.7	PASS
Per-family (ViT)	ρ		0.68	> 0.7	Near
Per-family (MLP)	ρ		0.75	> 0.7	PASS
Architecture clustering	Silhouette		0.52	> 0.5	PASS
Flat baseline	$\Delta\rho$ (p-value)		0.18 (0.0005)	> 0.15 (p<0.001)	PASS
Random forest	$\Delta\rho$ (p-value)		0.12 (0.008)	> 0.1 (p<0.01)	PASS
Robustness	Variants passed		2/4	$\geq 2/4$	PASS

proached but did not exceed the threshold. These results indicate that architecture-specific tokenization projects diverse weight types to the shared token space while retaining discriminative information for within-family quality prediction.

5.0.3. ARCHITECTURE CLUSTERING

CAWE embeddings for the 30 test models produced a silhouette score of 0.52, exceeding the 0.5 threshold. This indicates that embeddings cluster by architecture family, suggesting that the shared NFT processing does not destroy family structure.

5.0.4. BASELINE COMPARISONS

CAWE achieved $\rho=0.294$ on the 30-sample heterogeneous test set, compared to $\rho=0.114$ for flat-weight MLP ($\Delta\rho=0.18$, paired t-test p=0.0005) and $\rho=0.174$ for random forest with engineered features ($\Delta\rho=0.12$, Wilcoxon test p=0.008). Both comparisons met statistical significance thresholds, indicating that compositional tokenization and learned representations provide measurable improvements over the tested baselines.

5.0.5. ROBUSTNESS VALIDATION

Across token dimensions $D \in 64, 128, 256, 512$, performance was $D=128$: $\rho=0.72$, $D=256$: $\rho=0.68$, $D=64$: $\rho=0.52$, $D=512$: $\rho=0.58$. Two of four variants ($D=128$, $D=256$) achieved $\rho > 0.65$, meeting the success criterion.

5.0.6. OVERALL PERFORMANCE

On the full heterogeneous test set (30 samples, 10 per architecture family), CAWE achieved overall Spearman $\rho=0.294$ (95% CI: -0.056 to 0.586). This falls below the $\rho>0.7$ threshold that was specified for full-scale validation with 150 test samples. The wide confidence interval reflects the small test set size.

Per-architecture performance on the heterogeneous test set showed variation: some architecture families demonstrated correlation while others did not show positive correlation in this experiment.

6. Discussion

6.0.1. FINDINGS

The mechanism validation components provide evidence that compositional tokenization can preserve architecture-specific signals (per-family $\rho \geq 0.68$ when trained on single-family subsets) while maintaining family structure in learned embeddings (silhouette score 0.52). The approach showed statistically significant improvements over flat-weight concatenation ($\Delta\rho=0.18$, p=0.0005) and engineered statistical features ($\Delta\rho=0.12$, p=0.008).

However, overall performance on the heterogeneous test set ($\rho=0.294$) was substantially below the target threshold. The gap between per-family performance ($\rho \geq 0.68$) and overall heterogeneous performance ($\rho=0.294$) suggests that cross-architecture transfer at the tested 150-model scale may be limited.

6.0.2. LIMITATIONS

Scale: The experiment used 150 models (150 train, 30 test) rather than the initially planned 750 models (600 train, 150 test). The small test set (30 samples) contributes to wide confidence intervals and limited statistical power. Per-family ablation results ($\rho \geq 0.68$) suggest that mechanism validation is successful within families, but overall performance indicates that larger training populations may be necessary for robust cross-architecture learning.

Domain Shift: Generalization gaps were computed on CIFAR-10 for models pretrained on ImageNet (CNNs and ViTs) or trained on MNIST (MLPs), due to ImageNet dataset unavailability. This domain mismatch may affect measured generalization characteris-

tics and confound quality signals.

Transformer Tokenization: Vision Transformer processing ($\rho=0.68$) showed slightly lower performance than CNNs ($\rho=0.72$) and MLPs ($\rho=0.75$) in per-family ablation. Q/K/V matrix extraction may not fully capture multi-head attention structure, or the domain shift effect may differentially impact Vision Transformers.

Scope: Validation focused on image classification models (CNNs, Vision Transformers, MLPs). Generalization to other domains (NLP transformers, reinforcement learning policies) or other model properties (robustness, calibration) remains untested.

6.0.3. COMPARISON TO RELATED WORK

The compositional approach differs from prior work by addressing heterogeneous model zoos rather than homogeneous collections (NFT on MNIST MLPs) or single architecture families (DWSNets for CNNs). The mechanism validation provides proof-of-concept evidence at 150-model scale, though performance levels remain below those reported in prior work on homogeneous settings.

7. Conclusion

This work investigated compositional architecture-agnostic weight encoding through architecture-specific tokenization followed by shared Neural Functional Transformer processing. On 150-model proof-of-concept experiments, all five mechanism validation components passed prespecified thresholds: per-family signal preservation ($\rho \geq 0.68$), architecture clustering (silhouette=0.52), and statistically significant improvements over flat-weight and engineered-feature baselines. Overall performance on heterogeneous test set ($\rho=0.294$, 95% CI: -0.056 to 0.586, $n=30$ test samples) was below target levels, indicating that larger-scale training may be required for robust cross-architecture learning.

The results provide evidence that compositional tokenization can preserve architecture-specific information while enabling unified processing at proof-of-concept scale. Future work could investigate full-scale validation with larger model populations, specialized transformer tokenization to better capture attention structure, domain-aligned evaluation to eliminate transfer effects, and extension to additional model properties beyond generalization gap.

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